

Supplement to “Beyond Single-Positive: A Perturbed Fenchel-Young Loss for Top- K Recommendation”

1 Universality of the Perturbed Fenchel-Young Loss

In this section, we demonstrate that our proposed Perturbed Fenchel-Young (FY) framework acts as a universal umbrella that subsumes standard recommendation surrogate losses as special cases. Recall the general definition of the FY loss:

$$\mathcal{L}_{\text{FY}}(\mathbf{s}_u, \mathbf{y}_u) = W(\mathbf{s}_u; \mathcal{C}) - \langle \mathbf{s}_u, \mathbf{y}_u \rangle + \Omega(\mathbf{y}_u), \quad (1)$$

where $W(\mathbf{s}_u; \mathcal{C})$ is the surplus function over the discrete choice space $\mathcal{C}$. For deterministic ground-truth labels (i.e., elements in $\mathbf{y}_u \in \{0, 1\}$), the generalized entropy term $\Omega(\mathbf{y}_u) = \sum y \log y$ strictly evaluates to zero and is thus omitted. Below, we recover the BCE, BPR, and Softmax losses by simply varying the choice space $\mathcal{C}$ under standard Gumbel perturbations.

1.1 Recovering Pointwise BCE Loss

For pointwise binary classification of a single item with predicted score s and label $y \in \{0, 1\}$, the choice space is binary: $\mathcal{C} = \{0, 1\}$. Assuming standard Gumbel noise, the surplus function becomes the Softplus: $W(s) = \log(1 + \exp(s))$. Substituting this into Eq. (1) yields:

$$\mathcal{L}_{\text{FY}}(s, y) = \log(1 + \exp(s)) - y \cdot s, \quad (2)$$

which is algebraically identical to the standard Binary Cross-Entropy (BCE) loss: $-[y \log \sigma(s) + (1 - y) \log(1 - \sigma(s))]$.

1.2 Recovering Pairwise BPR Loss

For a positive item i and negative item j with scores $\mathbf{s} = [s_i, s_j]^\top$, the choice space forces the selection of exactly one item: $\mathcal{C} = \{\mathbf{c} \in \{0, 1\}^2 \mid c_1 + c_2 = 1\}$. Under i.i.d. Gumbel noise, $W(\mathbf{s}) = \log(\exp(s_i) + \exp(s_j))$. Given the ground-truth preference $\mathbf{y} = [1, 0]^\top$, the FY loss evaluates to:

$$\mathcal{L}_{\text{FY}}(\mathbf{s}, \mathbf{y}) = \log(\exp(s_i) + \exp(s_j)) - s_i = \log(1 + \exp(s_j - s_i)). \quad (3)$$

This formulation exactly matches the standard BPR loss: $-\log \sigma(s_i - s_j)$.

1.3 Recovering Listwise Softmax Loss

For listwise recommendation over M items, the conventional choice space assumes a one-hot selection: $\mathcal{C} = \{\mathbf{c} \in \{0, 1\}^M \mid \sum_{i=1}^M c_i = 1\}$. With i.i.d. Gumbel noise, the surplus function is the Log-Sum-Exp: $W(\mathbf{s}_u) = \log(\sum_{i=1}^M \exp(s_{ui}))$. For a target positive item t , the one-hot label $\mathbf{y}_u$ has $y_{ut} = 1$. The FY loss then yields:

$$\mathcal{L}_{\text{FY}}(\mathbf{s}_u, \mathbf{y}_u) = \log \left(\sum_{i=1}^M \exp(s_{ui}) \right) - s_{ut} = -\log \left(\frac{\exp(s_{ut})}{\sum_{i=1}^M \exp(s_{ui})} \right). \quad (4)$$

This perfectly recovers the standard categorical Softmax (or InfoNCE) loss.

Summary. By upgrading the choice space constraint from $\sum c_i = 1$ to $\sum c_i = P$, we naturally transition from these conventional single-positive paradigms to our proposed multi-positive Fenchel-Young framework.

2 Experiments Details

2.1 Evaluation Metrics for Ranking

The ultimate objective of the recommendation model is to optimize the top- K ranking quality, which is typically evaluated by discrete ranking metrics such as Precision@ K , Recall@ K , Normalized Discounted Cumulative Gain (NDCG@ K), and Mean Reciprocal Rank (MRR). Formally, we define the ranking operation as a permutation π_u obtained by sorting the predicted score vector $\mathbf{s}_u$ in descending order, where $\text{rank}(i|\mathbf{s}_u)$ denotes the rank position of item i (starting from 1).

Given a top- K recommendation list $\mathcal{R}_u^{(K)}$ generated by the model for user u , **Precision@ K** measures the proportion of retrieved items that are genuinely relevant to the user:

$$\text{Precision@}K = \frac{|\mathcal{R}_u^{(K)} \cap \mathcal{I}_u^+|}{K}. \quad (5)$$

Recall@ K , on the other hand, measures the proportion of all ground-truth relevant items that are successfully retrieved within the top- K list:

$$\text{Recall@}K = \frac{|\mathcal{R}_u^{(K)} \cap \mathcal{I}_u^+|}{|\mathcal{I}_u^+|}. \quad (6)$$

Beyond evaluating the unweighted sets, position-aware metrics are heavily emphasized, as users typically focus on the very top of the list. **NDCG@ K** assigns higher importance to relevant items ranked at higher positions with a

logarithmic discount:

$$\text{NDCG@}K = \frac{1}{\text{IDCG@}K} \sum_{i \in \mathcal{R}_u^{(K)}} \frac{y_{ui}}{\log_2(\text{rank}(i|\mathbf{s}_u) + 1)}, \quad (7)$$

where $\text{IDCG@}K$ is the ideal discounted cumulative gain obtained by a perfectly ordered ranking.

Similarly, **MRR** evaluates how early the model can successfully retrieve the *first* relevant item. For a specific user u , it evaluates the reciprocal rank of the highest-ranked positive item, and the overall MRR is averaged across all users $\mathcal{U}$:

$$\text{MRR} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \left(\max_{i \in \mathcal{I}_u^+} \frac{1}{\text{rank}(i|\mathbf{s}_u)} \right). \quad (8)$$

Since the sorting operation $\text{rank}(\cdot)$ fundamentally involves a discrete, non-differentiable mechanism (e.g., the argsort operator), none of the aforementioned metrics can be optimized directly via standard gradient descent. Consequently, this necessitates the design of continuous surrogate loss functions to effectively train deep recommendation models.

Evaluation Protocol. To assess the top- K ranking performance, we rank all unobserved items in the catalog for each user and report the average Precision@ K ($\text{P@}K$), Recall@ K ($\text{R@}K$), NDCG@ K ($\text{N@}K$), and MRR. We set the truncation length to $K = 20$ for all experiments.

2.2 Experimental Setup

For reproducibility, we provide an anonymous implementation of PFY: <https://anonymous.4open.science/r/PFY-E24F>.

In our training process, we utilize P -hot vectors to represent user-item interactions. The term “multi-hot” indicates that multiple positive items are considered for a given user, where P serves as a fixed hyperparameter denoting the exact number of positive samples drawn during each training step.

Datasets. We conduct experiments on four widely-used public benchmark datasets that vary across domains and sparsity levels: **Beauty**, **VideoGames**, **Gowalla**, and **Pinterest**. Following standard procedures, all datasets are pre-processed via 5-core filtering.

Table 1: Statistics of the datasets used in our experiments.

Dataset	Users	Items	Interactions	Density
Beauty	15,603	8,693	139,554	0.001029
VideoGames	24,303	10,672	231,780	0.000894
Gowalla	29,858	40,981	1,027,370	0.000840
Pinterest	55,187	9,916	1,028,553	0.001879

Backbone Architectures. To demonstrate the universality and model-agnostic

nature of our proposed loss, we evaluate it atop three representative recommendation architectures with varying complexities:

- **MF (Matrix Factorization) (Rendle et al., 2012):** A fundamental collaborative filtering model capturing linear user-item interactions.
- **LightGCN (He et al., 2020):** A highly prevalent graph neural network (GNN) that aggregates higher-order collaborative signals.
- **XSimGCL (Yu et al., 2023):** A state-of-the-art contrastive learning framework that enhances representation learning via cross-layer augmentations.

Baselines (Loss Functions). We compare our PFY against a comprehensive suite of recommendation losses, which can be categorized into: (1) *Conventional top-1 oriented losses*, including the pairwise **BPR** (Rendle et al., 2012) and the listwise **Softmax** (Wu et al., 2024b). (2) *Advanced Softmax variants*, including **BSL** (Wu et al., 2024a) and **PSL** (Yang et al., 2024). (3) *Metric-driven and smoothed ranking losses*, including **SmoothI** (Thonet et al., 2022) and **LLPAUC** (Shi et al., 2024).

Implementation Details. All models and loss functions are implemented in PyTorch. To ensure a fair comparison, the latent embedding dimension is fixed at $d = 64$ across all backbone architectures. We optimize the models using the Adam optimizer, tuning the learning rate based on validation performance over a predefined search space.

To ensure scalability and a fair comparison with baselines, the Top- P operation in PFY is not evaluated over the full item catalog. Instead, it is computed over a restricted candidate slate comprising the positive items and the uniformly sampled negative items, ensuring the time complexity strictly scales with the slate size rather than the full catalog.

For baselines requiring negative sampling, the number of negative samples is fixed at 128 per instance. For our proposed Perturbed Fenchel-Young (PFY) loss, we generate $N_s = 32$ independent Monte Carlo forward perturbations to robustly estimate the marginal probabilities. We tune the positive cardinality P within $\{1, 2, 3, 4, 5\}$, and the Gumbel perturbation temperature τ is selected from $\{0.05, 0.10, 0.20\}$. Additional hyperparameter settings and training details are provided in the supplementary material.

Table 2: Overall recommendation performance in terms of Precision@20 and Recall@20. All reported results are test means over five formal seeds.

Model	Loss	Beauty		VideoGames		Gowalla		Pinterest	
		P@20	R@20	P@20	R@20	P@20	R@20	P@20	R@20
MF	BPR	0.0124	0.1085	0.0044	0.0886	0.0298	0.1039	0.0285	0.1080
	BSL	0.0148	0.1236	0.0063	0.1264	0.0439	0.1448	0.0375	0.1425
	Softmax	0.0139	0.1223	0.0062	0.1239	0.0384	0.1280	0.0372	0.1415
	SmoothI	0.0144	0.1240	0.0057	0.1134	0.0428	0.1402	0.0379	0.1439
	PSL	0.0151	0.1260	0.0063	0.1256	0.0442	0.1443	0.0289	0.1097
	LLPAUC	0.0134	0.1071	0.0057	0.1140	0.0472	0.1575	0.0325	0.1235
	BCE	0.0122	0.1087	0.0043	0.0856	0.0311	0.1084	0.0282	0.1065
	PFY(ours)	0.0163	0.1387	0.0066	0.1328	0.0463	0.1525	0.0381	0.1446
LightGCN	BPR	0.0124	0.1116	0.0042	0.0843	0.0266	0.0874	0.0251	0.0951
	BSL	0.0154	0.1308	0.0062	0.1248	0.0462	0.1503	0.0390	0.1480
	Softmax	0.0156	0.1329	0.0064	0.1278	0.0374	0.1239	0.0348	0.1325
	SmoothI	0.0152	0.1308	0.0048	0.0951	0.0448	0.1448	0.0395	0.1501
	PSL	0.0154	0.1292	0.0063	0.1263	0.0419	0.1380	0.0306	0.1157
	LLPAUC	0.0148	0.1317	0.0053	0.1066	0.0298	0.1026	0.0206	0.0773
	BCE	0.0123	0.1110	0.0041	0.0828	0.0239	0.0858	0.0252	0.0949
	PFY(ours)	0.0164	0.1387	0.0067	0.1333	0.0472	0.1549	0.0412	0.1569
XSimGCL	BPR	0.0131	0.1080	0.0052	0.1048	0.0310	0.1081	0.0335	0.1274
	BSL	0.0140	0.1125	0.0063	0.1268	0.0458	0.1490	0.0416	0.1581
	Softmax	0.0153	0.1295	0.0063	0.1263	0.0393	0.1311	0.0417	0.1584
	SmoothI	0.0127	0.1005	0.0048	0.0960	0.0393	0.1253	0.0327	0.1244
	PSL	0.0153	0.1266	0.0063	0.1269	0.0419	0.1383	0.0297	0.1114
	LLPAUC	0.0156	0.1331	0.0061	0.1213	0.0378	0.1307	0.0375	0.1428
	BCE	0.0146	0.1216	0.0060	0.1190	0.0359	0.1253	0.0352	0.1336
	PFY(ours)	0.0162	0.1355	0.0065	0.1306	0.0460	0.1510	0.0422	0.1604

Table 3: Overall recommendation performance in terms of MRR@20 and NDCG@20. All reported results are test means over five formal seeds.

Model	Loss	Beauty		VideoGames		Gowalla		Pinterest	
		MRR	N@20	MRR	N@20	MRR	N@20	MRR	N@20
MF	BPR	0.0312	0.0591	0.0212	0.0357	0.0213	0.0760	0.0239	0.0720
	BSL	0.0399	0.0703	0.0314	0.0519	0.0330	0.1100	0.0329	0.0970
	Softmax	0.0352	0.0665	0.0307	0.0508	0.0278	0.0956	0.0320	0.0954
	SmoothI	0.0401	0.0712	0.0302	0.0482	0.0328	0.1078	0.0334	0.0981
	PSL	0.0399	0.0710	0.0316	0.0519	0.0339	0.1109	0.0246	0.0737
	LLPAUC	0.0353	0.0614	0.0305	0.0486	0.0427	0.1312	0.0280	0.0832
	BCE	0.0309	0.0587	0.0206	0.0345	0.0231	0.0812	0.0245	0.0723
	PFY(ours)	0.0440	0.0786	0.0331	0.0546	0.0360	0.1182	0.0334	0.0985
LightGCN	BPR	0.0307	0.0596	0.0197	0.0335	0.0180	0.0630	0.0214	0.0639
	BSL	0.0406	0.0730	0.0311	0.0513	0.0349	0.1149	0.0344	0.1010
	Softmax	0.0415	0.0746	0.0317	0.0524	0.0261	0.0911	0.0287	0.0874
	SmoothI	0.0416	0.0742	0.0232	0.0386	0.0338	0.1108	0.0351	0.1028
	PSL	0.0413	0.0731	0.0317	0.0521	0.0304	0.1034	0.0263	0.0781
	LLPAUC	0.0381	0.0717	0.0253	0.0427	0.0209	0.0746	0.0151	0.0485
	BCE	0.0309	0.0596	0.0194	0.0330	0.0168	0.0617	0.0217	0.0641
	PFY(ours)	0.0441	0.0786	0.0334	0.0549	0.0367	0.1202	0.0357	0.1061
XSimGCL	BPR	0.0326	0.0589	0.0247	0.0419	0.0199	0.0746	0.0277	0.0842
	BSL	0.0375	0.0645	0.0313	0.0519	0.0343	0.1135	0.0364	0.1075
	Softmax	0.0394	0.0716	0.0311	0.0516	0.0272	0.0953	0.0363	0.1074
	SmoothI	0.0332	0.0570	0.0235	0.0391	0.0303	0.0981	0.0274	0.0828
	PSL	0.0403	0.0713	0.0315	0.0520	0.0301	0.1027	0.0257	0.0757
	LLPAUC	0.0419	0.0754	0.0302	0.0498	0.0300	0.1012	0.0315	0.0951
	BCE	0.0370	0.0671	0.0289	0.0483	0.0244	0.0890	0.0295	0.0888
	PFY(ours)	0.0429	0.0764	0.0326	0.0537	0.0352	0.1161	0.0369	0.1090

Table 4: Effect of positive cardinality P on PFY with MF. Results are averaged over five seeds.

Dataset	P	P@20	R@20	N@20	MRR
Beauty	1	0.01495	0.12678	0.07199	0.04038
	2	0.01558	0.13210	0.07470	0.04171
	3	0.01562	0.13268	0.07513	0.04205
	4	0.01609	0.13647	0.07757	0.04350
	5	0.01633	0.13873	0.07859	0.04400
VideoGames	1	0.00619	0.12388	0.05079	0.03075
	2	0.00634	0.12689	0.05215	0.03162
	3	0.00643	0.12860	0.05278	0.03196
	4	0.00654	0.13072	0.05387	0.03278
	5	0.00664	0.13281	0.05461	0.03313
Gowalla	1	0.04296	0.14319	0.10966	0.03299
	2	0.04464	0.14796	0.11372	0.03439
	3	0.04544	0.15003	0.11592	0.03525
	4	0.04588	0.15133	0.11716	0.03570
	5	0.04630	0.15253	0.11816	0.03602
Pinterest	1	0.03418	0.12999	0.08792	0.02959
	2	0.03616	0.13741	0.09321	0.03149
	3	0.03694	0.14039	0.09552	0.03237
	4	0.03753	0.14275	0.09720	0.03298
	5	0.03807	0.14463	0.09848	0.03342

Table 5: Effect of positive cardinality P on PFY with LightGCN. Results are averaged over five seeds.

Dataset	P	P@20	R@20	N@20	MRR
Beauty	1	0.01564	0.13243	0.07437	0.04146
	2	0.01585	0.13432	0.07566	0.04228
	3	0.01589	0.13460	0.07613	0.04260
	4	0.01617	0.13714	0.07734	0.04323
	5	0.01638	0.13866	0.07863	0.04412
VideoGames	1	0.00640	0.12799	0.05251	0.03178
	2	0.00642	0.12838	0.05266	0.03187
	3	0.00651	0.13019	0.05346	0.03240
	4	0.00660	0.13196	0.05423	0.03288
	5	0.00667	0.13333	0.05493	0.03342
Gowalla	1	0.04552	0.14974	0.11531	0.03503
	2	0.04653	0.15271	0.11820	0.03604
	3	0.04689	0.15395	0.11911	0.03630
	4	0.04699	0.15424	0.11969	0.03658
	5	0.04723	0.15487	0.12021	0.03674
Pinterest	1	0.04015	0.15265	0.10248	0.03424
	2	0.04065	0.15454	0.10439	0.03513
	3	0.04093	0.15567	0.10522	0.03539
	4	0.04115	0.15634	0.10565	0.03555
	5	0.04124	0.15687	0.10606	0.03568

Table 6: Effect of positive cardinality P on PFY with XSimGCL. Results are averaged over five seeds.

Dataset	P	P@20	R@20	N@20	M@20
Beauty	1	0.01478	0.12174	0.06802	0.03829
	2	0.01535	0.12727	0.07180	0.04054
	3	0.01560	0.13020	0.07330	0.04116
	4	0.01597	0.13373	0.07539	0.04246
	5	0.01618	0.13549	0.07643	0.04292
VideoGames	1	0.00604	0.12082	0.04895	0.02929
	2	0.00624	0.12482	0.05093	0.03068
	3	0.00633	0.12665	0.05179	0.03126
	4	0.00646	0.12928	0.05297	0.03204
	5	0.00653	0.13063	0.05370	0.03258
Gowalla	1	0.04248	0.13981	0.10462	0.03093
	2	0.04398	0.14437	0.10939	0.03277
	3	0.04506	0.14779	0.11272	0.03396
	4	0.04576	0.15003	0.11496	0.03478
	5	0.04602	0.15101	0.11608	0.03521
Pinterest	1	0.04084	0.15504	0.10470	0.03526
	2	0.04180	0.15870	0.10759	0.03635
	3	0.04205	0.15981	0.10845	0.03667
	4	0.04217	0.16023	0.10902	0.03692
	5	0.04224	0.16044	0.10902	0.03694

Figure 1: Sensitivity analysis of positive cardinality P on Precision@20 across datasets and backbones.

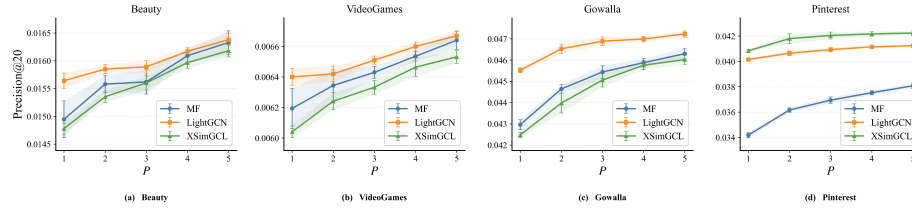
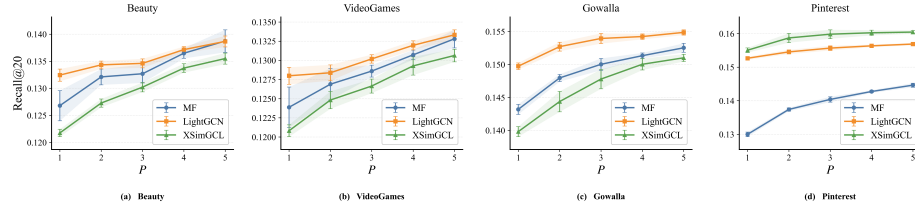


Figure 2: Sensitivity analysis of positive cardinality P on Recall@20 across datasets and backbones.



Training time. To evaluate the computational practicality of PFY-P5, we report the average single-run wall-clock training time over five random seeds on an NVIDIA GeForce RTX 4090 in Table 7. The results cover four representative datasets with different scales and three commonly used recommender-system backbones, namely MF, LightGCN, and XSimGCL. Overall, the training time remains within a practical range across all datasets and backbones. On the Beauty dataset, the training time is approximately 2.7–3.1 minutes, while it increases to approximately 5.1–6.2 minutes on VideoGames. For the larger Gowalla and Pinterest datasets, the training time remains below 20.3 minutes across all three backbones. These results demonstrate that PFY-P5 can be trained efficiently on a single GPU, while its computational cost remains manageable as the dataset scale increases.

Table 7: Average single-run wall-clock training time of PFY-P5 over five random seeds on an NVIDIA GeForce RTX 4090. Values are reported in seconds.

Dataset	MF	LightGCN	XSimGCL
Beauty	172.5	163.0	187.5
VideoGames	307.3	341.1	369.8
Gowalla	931.4	979.4	1143.5
Pinterest	869.5	1172.3	1211.3

Table 8: Hyperparameters to be searched for each method.

Method	Other hyperparameters
BPR	no other hyperparameters
Softmax	$\{\tau\}$
BSL	$\{\tau_+, \tau_-\}$
SmoothI	$\{\tau, \delta\}$
PSL	$\{\tau_*\}$
LLPAUC	$\{\alpha, \beta\}$
BCE	no other hyperparameters
Gumbel-PFY	$\{P, \epsilon\}$

Additional Implementation Details. The learning rate is selected based on validation performance from a predefined candidate set. For loss functions requiring negative sampling, the number of negative samples is uniformly fixed to 128 for each positive instance.

- **BPR/BCE:** No additional hyperparameters are introduced. The learning rate is searched in $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$.
- **Softmax:** The temperature τ is searched in $\{0.2, 0.5, 1.0\}$.
- **BSL:** The temperatures τ_+ and τ_- for positive and negative terms are searched in $\{0.05, 0.10, 0.15, 0.20\}$.
- **SmoothI:** The target rank is fixed to $K = 20$. The temperature τ and offset δ are searched in $\{0.02, 0.05, 0.10, 0.20\}$ and $\{0.10, 0.20\}$, respectively.
- **PSL:** The loss temperature τ_* is searched in $\{0.025, 0.05, 0.10\}$.
- **LLPAUC:** The optimization-region parameters α and β are searched in $\{0.7, 0.8, 0.9\}$ and $\{0.01, 0.02, 0.05\}$, respectively.
- **PFY:** The positive cardinality P is searched in $\{1, 2, 3, 4, 5\}$, and the Gumbel perturbation scale ϵ is searched in $\{0.05, 0.10, 0.20\}$. The number of Monte Carlo samples is fixed to $N_s = 32$.

Table 9: Optimal hyperparameters of IID setting on Beauty and VideoGames.

Model	Loss	Beauty				VideoGames			
		lr	wd	wl	others	lr	wd	wl	others
MF	BPR	10^{-2}	0	N/A	N/A	10^{-2}	0	N/A	N/A
	Softmax	10^{-2}	0	N/A	{0.5}	10^{-2}	0	N/A	{0.2}
	BSL	10^{-1}	0	N/A	{0.05, 0.15}	10^{-1}	0	N/A	{0.10, 0.20}
	SmoothI	10^{-1}	0	N/A	{0.20, 0.10}	10^{-1}	0	N/A	{0.20, 0.10}
	PSL	10^{-1}	0	N/A	{0.10}	10^{-1}	0	N/A	{0.10}
	LLPAUC	10^{-3}	0	N/A	{0.7, 0.05}	10^{-3}	0	N/A	{0.9, 0.02}
	BCE	10^{-2}	0	N/A	N/A	10^{-2}	0	N/A	N/A
	PFY	10^{-2}	0	N/A	{5, 0.20}	10^{-2}	0	N/A	{5, 0.20}
LightGCN	BPR	10^{-2}	0	N/A	N/A	10^{-2}	0	N/A	N/A
	Softmax	10^{-2}	0	N/A	{0.2}	10^{-2}	0	N/A	{0.2}
	BSL	10^{-1}	0	N/A	{0.15, 0.20}	10^{-1}	10^{-8}	N/A	{0.05, 0.20}
	SmoothI	10^{-1}	0	N/A	{0.20, 0.10}	10^{-1}	10^{-8}	N/A	{0.20, 0.10}
	PSL	10^{-2}	0	N/A	{0.10}	10^{-1}	0	N/A	{0.10}
	LLPAUC	10^{-2}	0	N/A	{0.7, 0.05}	10^{-2}	0	N/A	{0.7, 0.05}
	BCE	10^{-2}	0	N/A	N/A	10^{-2}	0	N/A	N/A
	PFY	10^{-2}	0	N/A	{5, 0.20}	10^{-2}	0	N/A	{5, 0.20}
XSimGCL	BPR	10^{-2}	0	0.05	N/A	10^{-2}	0	0.05	N/A
	Softmax	10^{-2}	0	0.05	{0.5}	10^{-3}	0	0.05	{0.2}
	BSL	10^{-3}	10^{-8}	0.20	{0.10, 0.10}	10^{-2}	0	0.20	{0.05, 0.20}
	SmoothI	10^{-3}	10^{-8}	0.02	{0.10, 0.10}	10^{-2}	0	0.02	{0.10, 0.10}
	PSL	10^{-1}	0	0.05	{0.10}	10^{-2}	0	0.05	{0.10}
	LLPAUC	10^{-4}	0	0.05	{0.7, 0.05}	10^{-4}	0	0.05	{0.7, 0.05}
	BCE	10^{-2}	0	0.05	N/A	10^{-2}	0	0.05	N/A
	PFY	10^{-2}	0	0.05	{5, 0.20}	10^{-3}	0	0.05	{5, 0.20}

Table 10: Optimal hyperparameters of IID setting on Gowalla and Pinterest.

Model	Loss	Gowalla				Pinterest			
		lr	wd	wl	others	lr	wd	wl	others
MF	BPR	10^{-2}	0	N/A	N/A	10^{-3}	0	N/A	N/A
	Softmax	10^{-2}	0	N/A	{0.2}	10^{-3}	0	N/A	{0.2}
	BSL	10^{-1}	0	N/A	{0.10, 0.10}	10^{-1}	0	N/A	{0.10, 0.10}
	SmoothI	10^{-1}	0	N/A	{0.10, 0.10}	10^{-1}	0	N/A	{0.10, 0.10}
	PSL	10^{-2}	0	N/A	{0.05}	10^{-2}	0	N/A	{0.025}
	LLPAUC	10^{-3}	0	N/A	{0.9, 0.01}	10^{-2}	0	N/A	{0.8, 0.01}
	BCE	10^{-2}	0	N/A	N/A	10^{-3}	0	N/A	N/A
	PFY	10^{-2}	0	N/A	{5, 0.10}	10^{-3}	0	N/A	{5, 0.10}
LightGCN	BPR	10^{-3}	0	N/A	N/A	10^{-3}	0	N/A	N/A
	Softmax	10^{-3}	0	N/A	{0.2}	10^{-3}	0	N/A	{0.2}
	BSL	10^{-1}	0	N/A	{0.10, 0.10}	10^{-1}	0	N/A	{0.10, 0.10}
	SmoothI	10^{-1}	0	N/A	{0.10, 0.10}	10^{-1}	0	N/A	{0.10, 0.10}
	PSL	10^{-2}	0	N/A	{0.10}	10^{-2}	0	N/A	{0.025}
	LLPAUC	10^{-2}	0	N/A	{0.7, 0.05}	10^{-4}	0	N/A	{0.8, 0.01}
	BCE	10^{-3}	0	N/A	N/A	10^{-3}	0	N/A	N/A
	PFY	10^{-2}	0	N/A	{5, 0.10}	10^{-3}	0	N/A	{5, 0.10}
XSimGCL	BPR	10^{-3}	0	0.05	N/A	10^{-3}	0	0.10	N/A
	Softmax	10^{-3}	0	0.20	{0.2}	10^{-3}	0	0.05	{0.2}
	BSL	10^{-2}	0	0.20	{0.05, 0.10}	10^{-2}	0	0.20	{0.05, 0.20}
	SmoothI	10^{-2}	0	0.10	{0.05, 0.10}	10^{-1}	0	0.10	{0.10, 0.10}
	PSL	10^{-2}	0	0.10	{0.10}	10^{-2}	0	0.10	{0.025}
	LLPAUC	10^{-4}	0	0.20	{0.7, 0.05}	10^{-3}	0	0.10	{0.8, 0.01}
	BCE	10^{-3}	0	0.05	N/A	10^{-3}	0	0.10	N/A
	PFY	10^{-2}	0	0.05	{5, 0.10}	10^{-3}	0	0.05	{5, 0.20}

References

- Krisztian Balog and ChengXiang Zhai. 2023. User simulation for evaluating information access systems. In *Proceedings of the Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region*. 302–305.
- Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yongdong Zhang, and Meng Wang. 2020. Lightgcn: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval*. 639–648.
- Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. 2012. BPR: Bayesian personalized ranking from implicit feedback. *arXiv preprint arXiv:1205.2618* (2012).
- Wentao Shi, Chenxu Wang, Fuli Feng, Yang Zhang, Wenjie Wang, Junkang Wu, and Xiangnan He. 2024. Lower-left partial auc: An effective and efficient optimization metric for recommendation. In *Proceedings of the ACM Web Conference 2024*. 3253–3264.
- Thibaut Thonet, Yagmur Gizem Cinar, Eric Gaussier, Minghan Li, and Jean-Michel Renders. 2022. Listwise learning to rank based on approximate rank indicators. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 8494–8502.
- Junkang Wu, Jiawei Chen, Jiancan Wu, Wentao Shi, Jizhi Zhang, and Xiang Wang. 2024a. Bsl: Understanding and improving softmax loss for recommendation. In *2024 IEEE 40th International Conference on Data Engineering (ICDE)*. IEEE, 816–830.
- Jiancan Wu, Xiang Wang, Xingyu Gao, Jiawei Chen, Hongcheng Fu, and Tianyu Qiu. 2024b. On the effectiveness of sampled softmax loss for item recommendation. *ACM Transactions on Information Systems* 42, 4 (2024), 1–26.
- Weiqin Yang, Jiawei Chen, Xin Xin, Sheng Zhou, Binbin Hu, Yan Feng, Chun Chen, and Can Wang. 2024. Psl: Rethinking and improving softmax loss from pairwise perspective for recommendation. *Advances in Neural Information Processing Systems* 37 (2024), 120974–121006.
- Junliang Yu, Xin Xia, Tong Chen, Lizhen Cui, Nguyen Quoc Viet Hung, and Hongzhi Yin. 2023. XSimGCL: Towards extremely simple graph contrastive learning for recommendation. *IEEE Transactions on Knowledge and Data Engineering* 36, 2 (2023), 913–926.